

S-Band RHCP Patch Antenna

Simulation Report

FR4 Substrate · 2.42 GHz · HFSS 2025 R2.4

1. Document Control

Version	Date	Description
V0	2026-06-07	Initial document

2. Simulation Environment

Item	Value	Notes
Simulation Software	Ansys HFSS 2025 R2.4	
Solver Type	Frequency Domain FEM	Adaptive mesh refinement
Frequency Sweep Type	Discrete	Discrete used for far-field AR vs freq
Adaptive Convergence	$\Delta S < 0.02$	
Radiation Boundary	Radiation (PML equivalent)	Air box dimensions: 300*300*200 mm
Port Type	Coaxial Wave Port	Terminal S-parameter solution Copper pin, Teflon based dielectric Inner radius = 0.635 mm Outer radius = 2.065 mm

3. Design Parameters

3.1 Substrate

Parameter	Value	Notes
Material	Rogers RO4350	
Relative Permittivity (ϵ_r)	4.4	HFSS value
Loss Tangent ($\tan \delta$)	0.02	HFSS value
Substrate Thickness (h)	1.52 mm	JLC offers "total thickness" of 1.6mm including Cu
Substrate Dimensions	100 × 100 mm	Ground plane / substrate extent
Copper Cladding Thickness	34.8 μm (1 oz)	

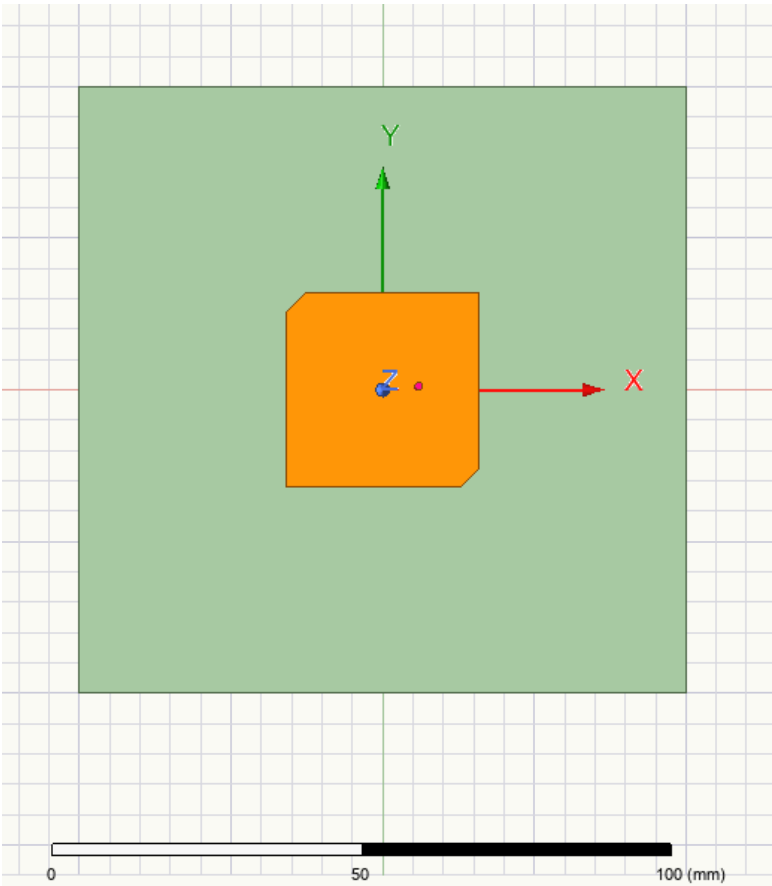
3.2 Patch Geometry

Parameter	Value	Notes
Patch Shape	Square	Corner-truncated for CP
Patch Side Length (L)	29 × 29 mm	
Corner Truncation (c)	3.3 mm	Leg of removed right-isosceles triangle
Truncated Corners	Top-Left & Bottom-Right	Produces RHCP
Intact Corners	Top-Right & Bottom-Left	

3.3 Feed

Parameter	Value	Notes
Feed Type	Coaxial Probe	SMA centre pin, 1.27 mm dia.
Feed X-Offset (feedx)	5.9 mm	From patch centre, +X direction
Feed Y-Offset (feedy)	0.6 mm	From patch centre, +Y direction
Feed Impedance	50 Ω	
Feed Axis	Off-diagonal	feedx ≠ feedy ensures both modes excited

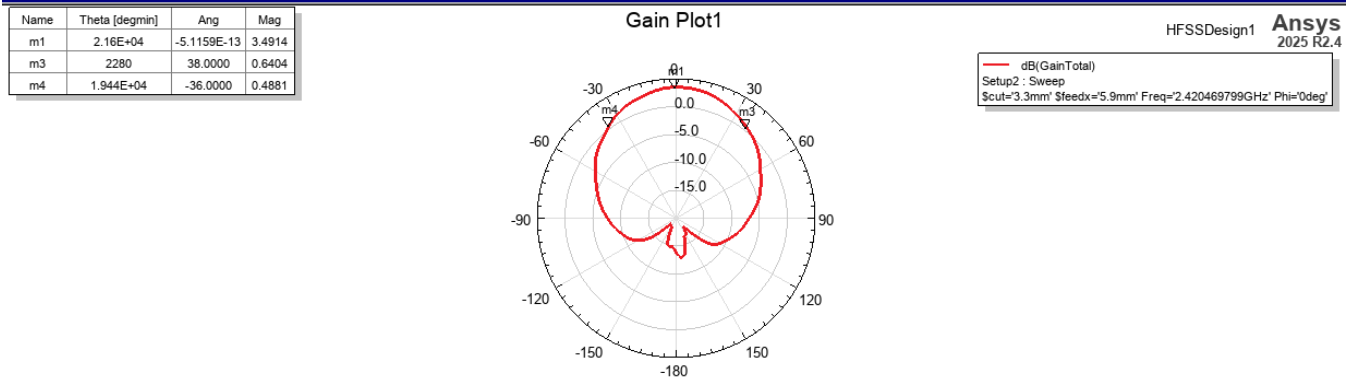
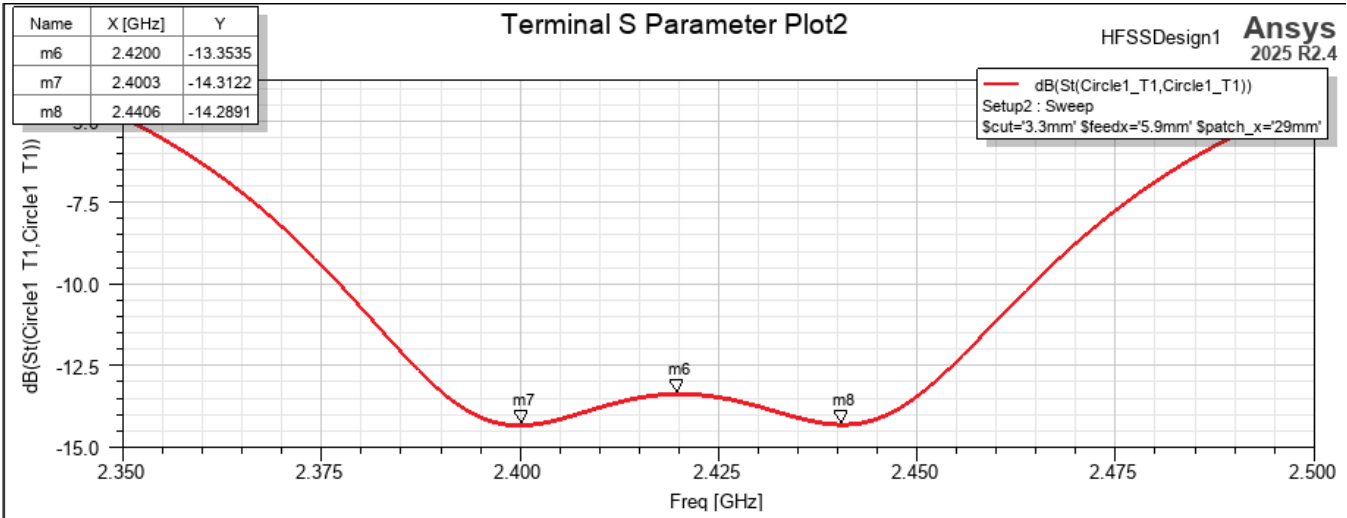
*image not to scale



4. Simulation Results

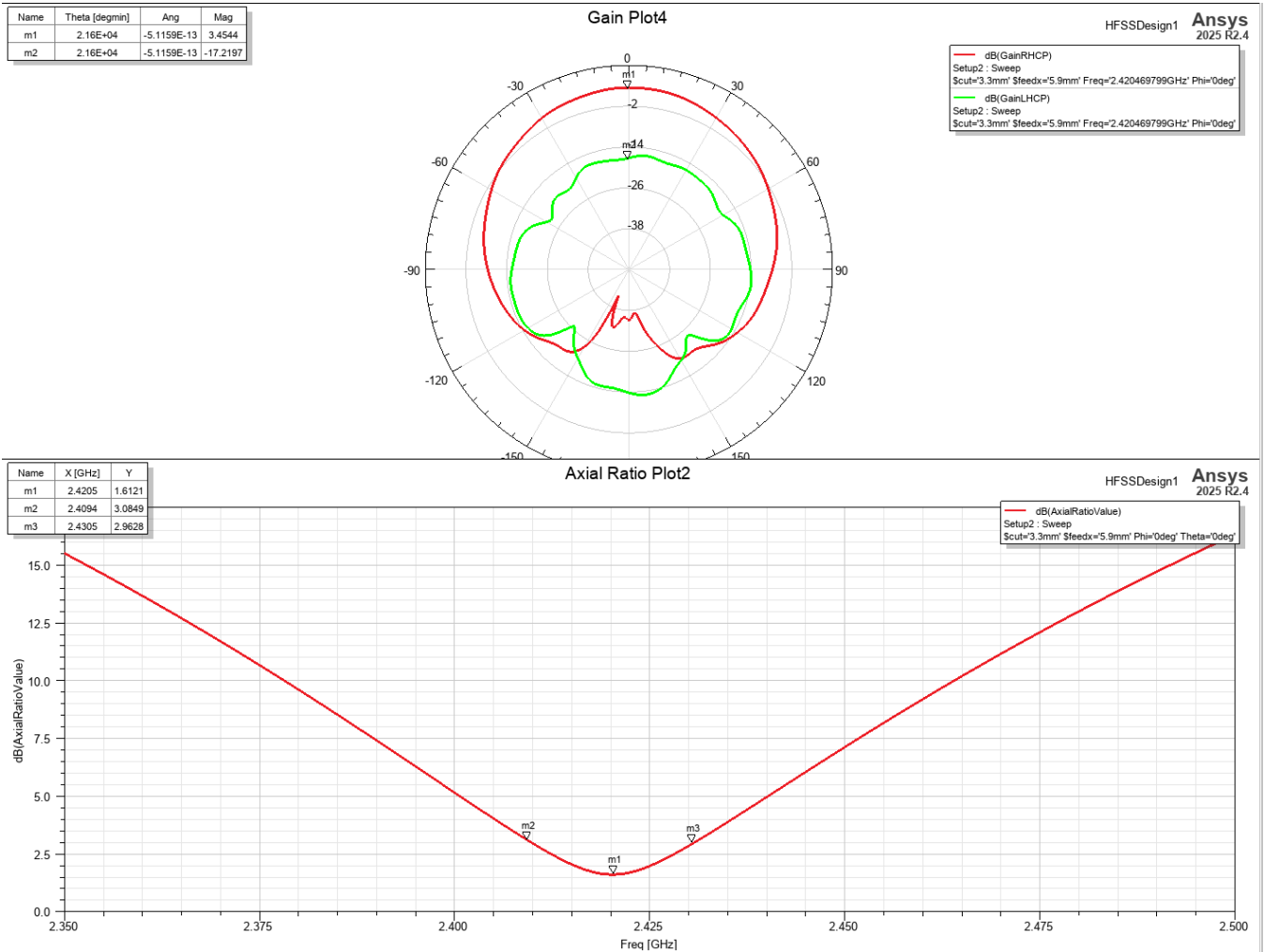
4.1 Impedance — S11

Parameter	Result	Notes
Centre Frequency (f0)	2.42 GHz	Demonstration purpose, will need to retune post frequency allocation
Lower Resonance (f1)	2.4003 GHz	First split mode dip
Upper Resonance (f2)	2.4406 GHz	Second split mode dip
Mode Split (f2 – f1)	40.3 MHz	
S11 at f1	-14 dB	
S11 at f2	-14 dB	
Dip Depth Balance	0 dB	
S11 at f0	-13 dB	
Simulated Q Factor	≈ 60	$Q = f_0 / \text{split} = 2420 / 40.3$

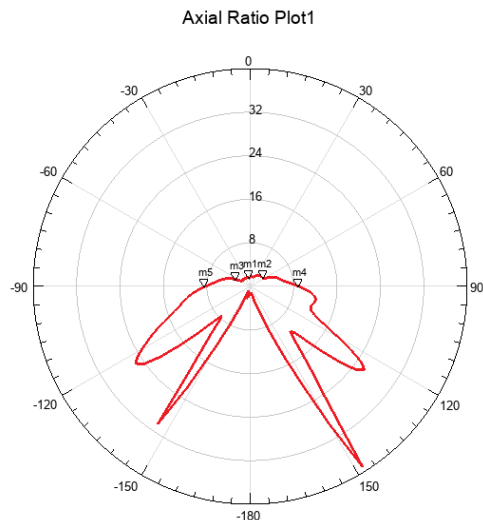


4.2 Radiation Performance

Parameter	Result	Notes
Polarisation Sense	RHCP	Confirmed from GainRHCP >> GainLHCP
GainRHCP at Boresight	+3.5 dBi	$\theta = 0^\circ, \varphi = 0^\circ, f = 2.42 \text{ GHz}$
GainLHCP at Boresight	-17 dBi	Cross-polarisation component
Cross-Pol Discrimination (XPD)	20.5 dB	GainRHCP – GainLHCP at boresight
Axial Ratio at Boresight	1.6 dB	< 3 dB threshold
AR at $\theta = +90^\circ$ (horizon)	9 dB	Expected degradation off-boresight
AR at $\theta = -90^\circ$ (horizon)	8 dB	
AR Bandwidth (< 3 dB)	$\approx 30 \text{ MHz}$	Determine sufficiency post frequency allocation
AR 3dB Beamwidth	$\sim 128 \text{ deg}$	
Peak Gain	3.5 dB	
Half-Power Beamwidth	$\sim 72 \text{ deg}$	
Radiation Efficiency	41.5 %	Extract from HFSS efficiency report
Total Efficiency	39.6 %	System efficiency



Name	Theta [degmin]	Ang	Mag
m1	2.16E+04	-5.1159E-13	1.6121
m2	3600	60.0000	2.9905
m3	1.782E+04	-64.0000	2.8493
m4	5400	90.0000	8.9917
m5	1.62E+04	-90.0000	8.2571



HFSSDesign1

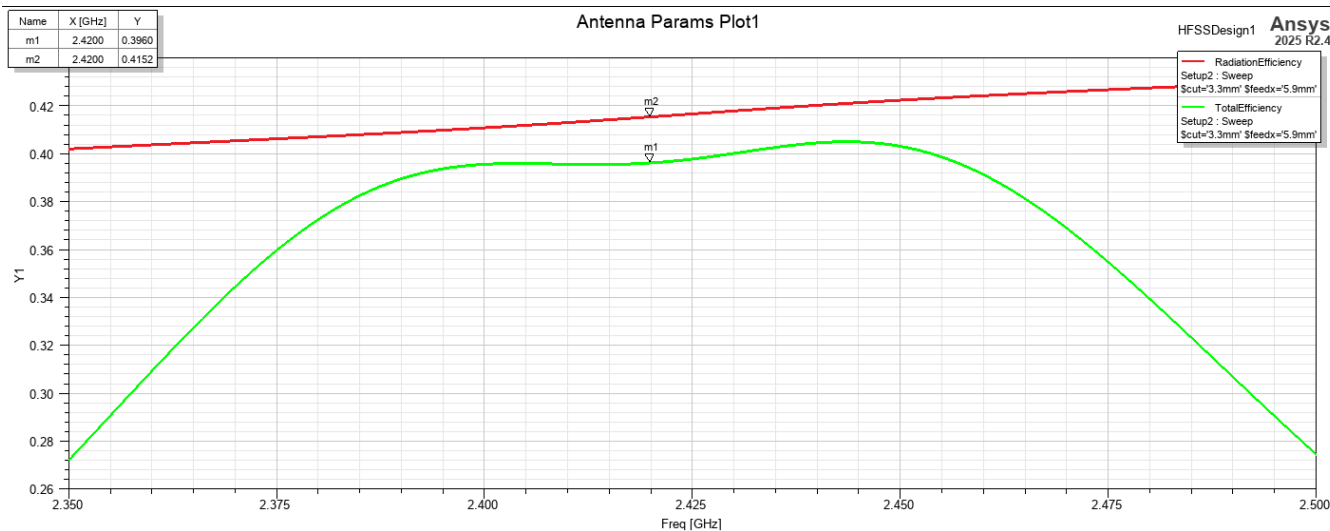
dB(AxialRatioValue)

Setup2 : Sweep

Scut=3.3mm' Steedx=5.9mm' Freq=2.420469799GHz' Phi=0deg'

Anslys

2025 R2.4



5. Link Budget Context

Parameter	Value	Notes
Application	CubeSat S-Band Downlink	Qubesat Mission
Target Frequency	2.42 GHz	Need retuning post frequency allocation
Orbit Altitude	LEO (TBD km)	Likely 400 km
Max Doppler Shift	±61 kHz	At 7.5 km/s orbital velocity
AR BW vs Doppler Margin	30 MHz / 0.12 MHz ≈ 266	Ample margin
Required Polarisation	RHCP	Standard ground station convention

6. Next Steps

#	Action	Owner	Status
1	Fabrication Review	TBD	Pending